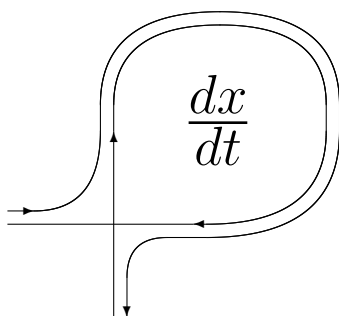




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Delay differential equations

Asymptotic stability of linear differential equations with diagonal delay via the Lyapunov-Krasovskii method

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Abstract. This paper investigates the asymptotic stability of the zero solution of some system of linear delay differential equations with diagonal delay. By constructing a Lyapunov functional and using the Lyapunov-Krasovskii method, sufficient criteria which ensure the asymptotic stability of the zero solution are established. The local asymptotic stability of a positive equilibrium of the Lotka-Volterra systems is investigated to demonstrate our results.

Key words and phrases: Asymptotic stability; Delay differential equations; stability criteria; Diagonal delay; Lyapunov-Krasovskii method

1. Introduction

For over five decades, the study of stability of the zero solution of delay differential equations with discrete and distributed delays in various forms has received tremendous attention in literature. Many researchers have studied linear and nonlinear delay differential equations of order one and higher orders, systems of linear differential equations with delay, and linear integro-differential

equation of Volterra type. Generally, the investigation of asymptotic stability of delay differential equations can be categorized into two distinct approaches; one approach deals with determining stability criteria or region of stability of the zero solution, while the other one is concerned with finding sufficient conditions for asymptotic stability of the zero solution. In order to obtain these conditions, authors have made use of the Lyapunov-Krasovskii functional method, the frequency domain method (see, e.g., [1,2,3,9,15,21,22,25,32] and the references cited therein), and the root analysis method (see, e.g. [4-8,10-14,16-20,23,24,26-31] and the references cited therein).

It is important to study delay differential equations since they often appear in applications in physics and engineering through mathematical models (e.g. population model, prey-predator models, neural network models, control theory models, to mention a few, see e.g. [7, 20] and the references cited therein).

Resently, Matsunaga [18] investigated a system of linear delay differential equations of the form:

$$\begin{aligned}x'(t) &= -ax(t - \tau) - by(t), \\y'(t) &= -cx(t) - ay(t - \tau),\end{aligned}\tag{1}$$

where a, b, c are real constants and τ a positive real constant, for the asymptotic stability of the zero solution. The author [18] obtained the necessary and sufficient criteria for the asymptotic stability of the system (1) using the root analysis method by completely investigating the roots of the associated characteristic equation

$$F(\lambda) \equiv \lambda^2 + 2a\lambda e^{-\lambda\tau} + a^2e^{-2\lambda\tau} - bc = 0.\tag{2}$$

The author obtained certain conditions on a, b, c and τ for which the roots ($\lambda = \alpha + i\beta$) of Eq. (2) have a negative real part.

Equations of the form (1) occur as the linearization of the population models of Lotka-Volterra type; for example, delayed prey-predator system in [29] and a delayed competitive system in [27]. Zhen and Ma [29] have considered a more general class (of system (1)) with distinct coefficients and distinct delays in each variable. In their investigation, they make use of the Lyapunov-Krasovskii method by constructing Lyapunov functionals to obtain sufficient conditions which guarantee the stability of the zero solution. But their results provide conditions which do not agree with Matsunaga [18] in application.

In the present investigation, we will study system (1) for asymptotic stability of

the zero solution using the Lyapunov-Krasovskii method, thereby constructing a more accurate Lyapunov functional (unlike in [29]) to obtain sufficient conditions which ensure asymptotic stability of the zero solution of system (1). In section 3, we will demonstrate how our result compare favorably with [18].

2. Asymptotic stability

Consider the system (1) in the form

$$\begin{aligned}\left(x(t) - a \int_{t-\tau}^t x(s)ds\right)' &= -ax(t) - by(t), \\ \left(y(t) - a \int_{t-\tau}^t y(s)ds\right)' &= -cx(t) - ay(t),\end{aligned}\tag{3}$$

where a, b, c are real constants and τ is a positive constant.

Let

$$A(t) = x(t) - a \int_{t-\tau}^t x(s)ds$$

and

$$B(t) = y(t) - a \int_{t-\tau}^t y(s)ds.$$

Theorem 1. *Consider the system (1) (equivalently system (3)), with a, b, c, τ real positive constants. Assume that*

$$(i) \ a^2 - bc > 0$$

$$(ii) \ 0 < \tau < a^{-1}.$$

Then the zero solution of system (1) (equivalently, system (3)) is asymptotically stable.

Proof of Theorem 1. The proof of Theorem 1 depends on the Lyapunov functional $W(t) = W(U)(t)$, which is defined for $U(t) = (x(t), y(t))$ as

$$W(t) = W_1(U)(t) + W_2(U)(t)\tag{4}$$

in which

$$\begin{aligned} W_1(U)(t) = & \left(\frac{2a}{b} - \frac{c}{a} \right) A^2(t) - 2A(t)B(t) + \left(\frac{b}{a} \right) B^2(t) \\ & + \lambda \int_{t-\tau}^t \int_s^t x^2(\theta) d\theta ds + \beta \int_{t-\tau}^t \int_s^t y^2(\theta) d\theta ds \end{aligned} \quad (5)$$

and

$$W_2(U)(t) = \left(\frac{c}{a} \right) A^2(t) - 2A(t)B(t) + \left(\frac{2a}{c} - \frac{b}{a} \right) B^2(t), \quad (6)$$

where λ and β are positive real constants that will be fixed to advantage later. To establish that (5) and (6) are positive-definite, we re-write $W_1(U)(t)$ and $W_2(U)(t)$ respectively, thus:

$$\begin{aligned} W_1(U)(t) = & \left(\frac{a}{b} - \frac{c}{a} \right) A^2(t) + \left(\sqrt{\frac{a}{b}} A(t) - \sqrt{\frac{b}{a}} B(t) \right)^2 \\ & + \lambda \int_{t-\tau}^t \int_s^t x^2(\theta) d\theta ds + \beta \int_{t-\tau}^t \int_s^t y^2(\theta) d\theta ds, \end{aligned} \quad (7)$$

and

$$W_2(U)(t) = \left(\frac{a}{c} - \frac{b}{a} \right) B^2(t) + \left(\sqrt{\frac{c}{a}} A(t) - \sqrt{\frac{a}{c}} B(t) \right)^2. \quad (8)$$

Obviously,

$$\begin{aligned} \lambda \int_{t-\tau}^t \int_s^t x^2(\theta) d\theta ds &> 0, \\ \beta \int_{t-\tau}^t \int_s^t y^2(\theta) d\theta ds &> 0, \end{aligned}$$

hence $W_1(U)(t)$ and $W_2(U)(t)$ now given by (7) and (8) respectively, are positive-definite.

Next, we obtain the derivative of $W(t)$ (in (4)) along the solution path of system

(3) as

$$\begin{aligned} W'(U)(t) &= -4a \left(\frac{a}{b} - \frac{c}{a} \right) x^2(t) - 4a \left(\frac{a}{c} - \frac{b}{a} \right) y^2(t) \\ &\quad + 4a^2 \left(\frac{a}{b} - \frac{c}{a} \right) x(t) \int_{t-\tau}^t x(s) ds + 4a^2 \left(\frac{a}{c} - \frac{b}{a} \right) y(t) \int_{t-\tau}^t y(s) ds \\ &\quad + \lambda \left\{ \tau x^2(t) - \int_{t-\tau}^t x^2(s) ds \right\} + \beta \left\{ \tau y^2(t) - \int_{t-\tau}^t y^2(s) ds \right\}. \end{aligned}$$

Since

$$\begin{aligned} x(t) \int_{t-\tau}^t x(s) ds &= \int_{t-\tau}^t x(t)x(s) ds \\ &\leq \frac{1}{2} \int_{t-\tau}^t x^2(t) ds + \frac{1}{2} \int_{t-\tau}^t x^2(s) ds \\ &= \frac{1}{2} \tau x^2(t) + \frac{1}{2} \int_{t-\tau}^t x^2(s) ds \end{aligned}$$

and

$$\begin{aligned} y(t) \int_{t-\tau}^t y(s) ds &= \int_{t-\tau}^t y(t)y(s) ds \\ &\leq \frac{1}{2} \int_{t-\tau}^t y^2(t) ds + \frac{1}{2} \int_{t-\tau}^t y^2(s) ds \\ &= \frac{1}{2} \tau y^2(t) + \frac{1}{2} \int_{t-\tau}^t y^2(s) ds, \end{aligned}$$

we must have that

$$\begin{aligned}
 W'(U)(t) \leq & -4a \left(\frac{a}{b} - \frac{c}{a} \right) x^2(t) - 4a \left(\frac{a}{c} - \frac{b}{a} \right) y^2(t) \\
 & + 2a^2 \left(\frac{a}{b} - \frac{c}{a} \right) \tau x^2(t) + 2a^2 \left(\frac{a}{c} - \frac{b}{a} \right) \tau y^2(t) \\
 & + 2a^2 \left(\frac{a}{b} - \frac{c}{a} \right) \int_{t-\tau}^t x^2(s) ds + 2a^2 \left(\frac{a}{c} - \frac{b}{a} \right) \int_{t-\tau}^t y^2(s) ds \\
 & + \lambda \left\{ \tau x^2(t) - \int_{t-\tau}^t x^2(s) ds \right\} + \beta \left\{ \tau y^2(t) - \int_{t-\tau}^t y^2(s) ds \right\}.
 \end{aligned}$$

Simplifying further, we get

$$\begin{aligned}
 W'(t) \leq & -2 \left\{ 2 \left(\frac{a^2 - bc}{b} \right) - \left[\frac{a(a^2 - bc)}{b} + \frac{1}{2} \lambda \right] \tau \right\} x^2(t) \\
 & - 2 \left\{ 2 \left(\frac{a^2 - bc}{c} \right) - \left[\frac{a(a^2 - bc)}{c} + \frac{1}{2} \beta \right] \tau \right\} y^2(t) \\
 & + \left\{ \frac{2a(a^2 - bc)}{b} - \lambda \right\} \int_{t-\tau}^t x^2(s) ds \\
 & + \left\{ \frac{2a(a^2 - bc)}{c} - \beta \right\} \int_{t-\tau}^t y^2(s) ds.
 \end{aligned}$$

If we set $\lambda = \frac{2a(a^2 - bc)}{b}$ and $\beta = \frac{2a(a^2 - bc)}{c}$ in the last inequality, we get

$$\begin{aligned}
 W'(t) \leq & -4 \left\{ \frac{(a^2 - bc)}{b} - \frac{a(a^2 - bc)}{b} \tau \right\} x^2(t) \\
 & - 4 \left\{ \frac{(a^2 - bc)}{c} - \frac{a(a^2 - bc)}{c} \tau \right\} y^2(t) \tag{9} \\
 \leq & -\gamma_1 x^2(t) - \gamma_2 y^2(t), \text{ whenever } \tau < a^{-1}.
 \end{aligned}$$

Denote $\gamma = \min\{\gamma_1, \gamma_2\}$, then (9) leads to

$$W(t) + \gamma \int_T^t [x^2(s) + y^2(s)] ds \leq W(T) \quad \text{for } t \geq T,$$

and this implies that $x^2(t) + y^2(t) \in L_1[T, \infty)$.

It is easy to see from (1) and $U(t) = (x(t), y(t))$ that $x^2(t) + y^2(t)$ is uniformly continuous and then, using Babalat's Lemma (See [9]), we can conclude that

$$\lim_{t \rightarrow \infty} \{x^2(t) + y^2(t)\} = 0.$$

Therefore the zero solution of (3) is asymptotically stable and this concludes the proof.

3. Applications

In this section, we investigate local stability of positive equilibrium of delayed Lotka-Volterra systems. The delayed competitive system

$$\begin{aligned} x'(t) &= x(t)[1 - x(t - \tau) - \alpha y(t)], \\ y'(t) &= y(t)[1 - \alpha x(t) - y(t - \tau)], \end{aligned} \tag{10}$$

where $\alpha > 0$ and $\tau > 0$ are real constants have been studied for stability (See [18, 29]). The system (10) has the unique positive equilibrium (x^*, y^*) , where $x^* = y^* = \frac{1}{\alpha + 1}$. When (10) is linearized around (x^*, y^*) , we get

$$\begin{aligned} x'(t) &= -\frac{1}{\alpha + 1}x(t - \tau) - \frac{\alpha}{\alpha + 1}y(t), \\ y'(t) &= -\frac{\alpha}{\alpha + 1}x(t) - \frac{1}{\alpha + 1}y(t - \tau). \end{aligned} \tag{11}$$

It is well-known that the local asymptotic stability of the zero solution of system (11) implies that the positive equilibrium (x^*, y^*) of (10) is locally asymptotically stable [18, Corollary 3.1], with

$$0 < \alpha < 1 \text{ and } \tau < \frac{1 + \alpha}{\sqrt{1 - \alpha^2}} \arccos \alpha. \tag{12}$$

Comparing systems (1) and (11), it is clear that $a = \frac{1}{\alpha + 1}, b = c = \frac{\alpha}{\alpha + 1}$. Therefore, applying Theorem 1 to system (11), we get

Corollary 1. *If*

$$0 < \alpha < 1 \text{ and } \tau < \alpha + 1, \quad (13)$$

then the positive equilibrium (x^, y^*) of system (10) is locally asymptotically stable.*

In the limit $\alpha \rightarrow 1-$, this estimate (i.e. (13)) agrees with (12), and it becomes worse as $\alpha \rightarrow 0+$: the difference tends to $-1 + \frac{\pi}{2} = 0.5707\dots$ For the considered example of $\alpha = \frac{1}{2}$, we have that $\tau < 1.5$ and $\tau < \frac{\pi}{\sqrt{3}} = 1.8137$, respectively.

Remark 1. In [29], Zhen and Ma have shown that under $\alpha = \frac{1}{2}$, the positive equilibrium $(\frac{2}{3}, \frac{2}{3})$ of (10) is locally asymptotically stable if $0 < \tau < \frac{1}{2}$. These are much worse conditions in the particular case $\alpha = 0.5$ in (11) compared to Corollary 1 (above) and [18, Corollary 3.1]. However, the authors were able to obtain a global stability result if $\tau < 0.257\dots$ in the particular case.

Future Work. Our next paper will focus on the open problem announced by Matsunaga [18]. We shall apply the Lyapunov-Krasovskii method to obtain results.

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